

Suryan Pinnoju

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EXPERIENCE

IkeGPS

August 2025 – May 2026

AI Engineering Contractor | *Ultralytics, MMDetection, MLFlow, Transformers*

Hyderabad, India

- Improved production keypoint detection accuracy by **7.6% over baseline achieving an accuracy of 81.3%**, by benchmarking and training models across YOLO, RT-DETR, and MMDetection frameworks
- Strengthened model robustness through 15+ validation and optimization iterations within Swin V1/V2 pipelines, including photometric distortion augmentation, hard-negative training, and finetuning for real-world variability.
- Fine-tuned YOLO architectures using mosaic augmentation, multiclass training, and 500+ manually curated annotations, significantly reducing generalization error across varied edge environments
- Worked with MLflow-based experiment tracking workflows for reproducible benchmarking and comparison across multiple detection architectures.
- Optimized hard-negative mining and evaluated semi-supervised student teacher learning with pseudolabeling on 25k - 35k unlabeled images for keypoint detection.

Navatej

March 2025 – August 2025

Research Intern | *Reinforcement Learning, OpenWorm*

Hyderabad, India

- Researched biologically inspired autonomous agents using multi-objective reinforcement learning in dynamic simulated environments.
- Developed an adaptive behavioral learning agent capable of discovering and reusing novel movement strategies.
- Designed persistent memory capabilities enabling prior environmental experiences to influence future decision-making.

Prodigal

March 2025 – August 2025

AI Intern (Part time) | *Python, Flask, APIs, OpenCV*

Hyderabad, India

- Led a cross-functional team of 4** to develop an AI-assisted video editing system, earning the **Silver Leadership Certificate** for exceptional project delivery.
- Developed a context adaptive custom attention-scoring framework using 4 computer vision engagement features.
- Architected rapid prototypes and experimentation pipelines, accelerating the transition from theoretical research to a functional software MVP.

PROJECTS

Multi Object Tracking

Pytorch, ByteTrack, Object Tracking, Segformers

- Developed a real-time vehicle detection and multi-object tracking pipeline using YOLO26 and ByteTrack for dashcam video
- Evaluated tracking performance on the KITTI MOT benchmark, achieving 64.5% MOTA and 76.9% IDF1 using standard MOT evaluation metrics.

LiDAR-Camera Sensor Fusion

Sensor Fusion, Pytorch, Open3D

- Built a multi-sensor perception pipeline on the KITTI dataset by projecting 3D LiDAR point clouds into camera images using calibration matrices
- Implemented SegFormer (MiT-B3) semantic segmentation and mapped pixel-wise scene labels onto LiDAR points to generate semantically colored point clouds.

EDUCATION

Mahindra University

B.Tech Computer Science

2021 –2025

Hyderabad, India